

Non-photorealistic compositing of the Spider-Verse

Showing the hand of the artist without painting every pixel

Marc-André Davignon
mdavignon@imageworks.com
Sony Pictures Imageworks
Montréal, QC, Canada

Bret St. Clair
bret@imageworks.com
Sony Pictures Imageworks
Los Angeles, CA, USA



©2021 CTMG, Inc. All Rights Reserved

Figure 1: Spider-Man: Across the Spider-Verse trailer image illustrating painterly effects.

ABSTRACT

The various worlds of Spider-Man: Across the Spider-Verse have wildly varying art styles and rules. Matching 2D concept art in an animated 3D world requires finding novel ways to marry the guiding hand of an artist with procedural tools. Building upon the Patcher Tool [et al. 2021], created for Spider-Man: Into the Spider-Verse, we designed and implemented a suite of new compositing nodes to apply painterly effects on 3D renders as a post-process. We discuss what they do, how they work, how they came to be and show results. We also share the development strategy used to design and build software solutions to a constantly evolving set of challenges that are actively used to produce artwork from very early prototypes.

ACM Reference Format:

Marc-André Davignon and Bret St. Clair. 2023. Non-photorealistic compositing of the Spider-Verse: Showing the hand of the artist without painting every pixel. In *Special Interest Group on Computer Graphics and Interactive Techniques Conference Talks (SIGGRAPH '23 Talks)*, August 06–10, 2023, Los Angeles, CA, USA. ACM, New York, NY, USA, 2 pages. <https://doi.org/10.1145/3587421.3595463>

1 INTRODUCTION

Inaccuracy, broken contours, off-surface paint, and layering are all common components of concept art that are not a natural for 3D renders. In the first movie the Patcher tool would introduce inaccuracy through noise patterns applied to a position pass in order to create tracking patches with individual ids. This allowed a post-process filter to collect color information from the render and recolor each patch. The second movie's varied looks required a much more painterly effect. Building upon the idea of a lower color-resolution patch we introduced shape to those patches in the form of scanned brush strokes distributed on geometry. In this talk, we will discuss custom tools built to distribute and render these brushes and how they evolved throughout production. The tools fall into three categories: Brush projection tools, 3D cards as brushes, and 3D curves as brushes.

2 BRUSH PATCHING

These tools are direct descendants of Patcher Tool [et al. 2021] they scale the world space of the incoming position pass to partition it

Permission to make digital or hard copies of part or all of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).

SIGGRAPH '23 Talks, August 06–10, 2023, Los Angeles, CA, USA

© 2023 Copyright held by the owner/author(s).

ACM ISBN 979-8-4007-0143-6/23/08.

<https://doi.org/10.1145/3587421.3595463>

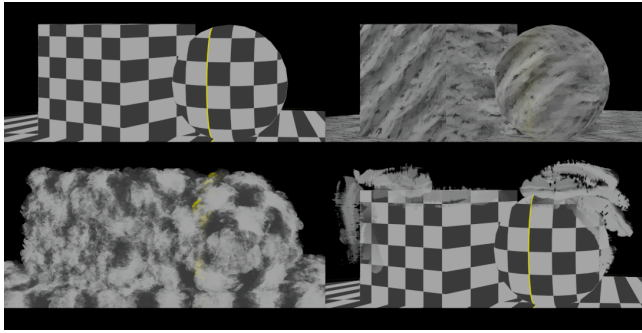


Figure 2: Cube and sphere with brush treatments. Original, Projection, 3D Cards, 3D Curves.

into rectangular regions that follow the contours of objects in the scene. Each pixel in that space is subtracted from its floor to generate a UV coordinate for each projection plane. The incoming brush texture is applied to each patch to be color-collected and filtered through the same mechanism as Patcher and we use a triplanar mix to generate the final color at each pixel. Users have control over UV space transforms and jitter within that space. Randomized transforms for offset, scale, rotation and opacity are essential to give a natural feel to the output. The brush texture is also randomly selected from a set of inputs. The result of this is a layer that has partial coverage over the surface. The patching tools render the number of layers requested by the user at different scales and offset them semi-randomly to provide surface coverage. The layers are composited together from larger to smaller strokes but randomly merge as over or under on a per-patch basis. These tools provide good coverage and the layering simulates the ordered application of larger brush strokes first effectively. They require no animation controls because they react to incoming passes and, as such, are very stable temporally. This also makes them react to lighting changes appropriately. The main limitation of these tools is that they cannot break contours and always stick to the surface.

3 3D CARDS AS BRUSHES

The need to break contours brought the focus towards positioning brushes on cards anchored to geometry at their center point in world position. Each brush has independent attributes for length, position, id, object id, age, draw frame, normal, three color samples, and transform values. These attributes flow through the node graph as a packed set of image channels with each pixel representing a card. Custom nodes that operate on the data channels are provided to the artist to introduce jitter, making it feel natural, along with controls to respond to image data in the destination frame. e.g. Now that brushes can extend off-surface, how far off-surface needs to be controlled so a z-threshold value is provided to determine if a pixel should grab a contribution from a card based on how far it is in z. Similar tests are provided to respond to orientation, object id, and color distance metrics. The render nodes can orient these cards to the normal on the object or orient them to be always camera facing, providing different looks. Because this approach involves the placement of objects, rather than per-pixel processing, distribution is a major challenge. A card needs to be at the same location on an

object on every frame. Scattering brushes on every frame creates crawling and popping issues and scattering brushes on a single frame creates gaps as new geometry is revealed. Gap filling options can work with static geometry but will not work with reference position due to the inability to project a p-ref value through camera at a different frame. Instead of using p-ref, we import point clouds with a unique id per point that have absolute positions in world space for the entire frame range for placement. These point clouds are generated in a 3D package by an upstream department.

4 3D CURVES AS BRUSHES

The need for hero strokes that have more complex shapes and that can follow geometry without having any point connected to that geometry brought us to write tools to hand draw Catmull-Rom curves in world space. The artist drawing a curve can set a reference point on any location in the image that will define a plane perpendicular to the normal at that location. Any control point that falls outside of where the position pass is defined will be placed on the reference plane. The same approach to packing point attributes in custom image channels as that used for 3D cards is used here, applied to control points, but with each point on a given curve sharing the same id. If we are dealing with static geometry, simple projection through the camera takes care of placement on destination frames. When strokes are meant to track deforming geometry, we leverage the point clouds generated for 3D card brushes by rigging curve control points to the closest point-cloud points. This is done, without user interaction, through a separate node. The curves are rendered as camera-facing strokes by evaluating the Catmull-Rom curve and displacing the points perpendicularly to the curve in opposite directions, creating a hull that respects the size attribute. These displaced curves are split into quads to generate UVs and then textured and colored using the same techniques used for 3D card brushes.

5 DEVELOPMENT PROCESS

Here, we highlight how we approached sharing development between look development artists on the production side and the compositing engineering team on the facility side. The need to empower experimentation without limiting iteration. How always having plans for new features or tools to explore, improving the user experience, optimizing performance and throughput, achieving stability, and avoiding regression is necessary, but cannot be treated linearly, since priorities can change daily, based on shot delivery timelines. We'll discuss the things that fared best and worse and what we learned from this extreme example.

6 CONCLUSION

We have outlined the tools and techniques for non-photorealistic rendering by compositors created for *Spider-Man: Across the Spider-Verse*. This work successfully provided a shortened pipeline with a fast iteration rate to deliver unique looks at a massive scale on a production schedule.

REFERENCES

- St. Clair et al. 2021. Patcher Tool. Patent No. US 11,195,308 B2, Filed Oct. 16th, 2019, Issued Dec. 7th, 2021.